Customer shopping behavior, from records to recommendations



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01 · THE QUESTION

10,000

PURCHASE RECORDS

Enough history to see patterns worth acting on.



10,000 purchases reveal a behavior story worth acting on

- The project analyzes spending patterns, customer segments, product preferences, and subscription behavior
- The goal is to turn transactional data into better marketing and sales decisions

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WHO × HOW

One record links customer context to shopping behavior.

The dataset connects who customers are with how they shop



10,000 records

Across 18 columns.



Demographics

Age, gender, location, and subscription status.



Purchases

Item, category, amount, season, size, and color.



Behavior

Discounts, purchase history, frequency, ratings, and shipping preference.

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Data preparation made the analysis consistent and reusable

A clean pipeline made every later question easier to trust.

01

Load

Loaded and explored the records with pandas.

02

Impute

Imputed 95 missing review ratings using the median for each product category.

03

Engineer

Standardized columns, created age groups, and engineered purchase-frequency days.

04

Reuse

Removed the redundant promo-code field and loaded the cleaned data into MySQL.

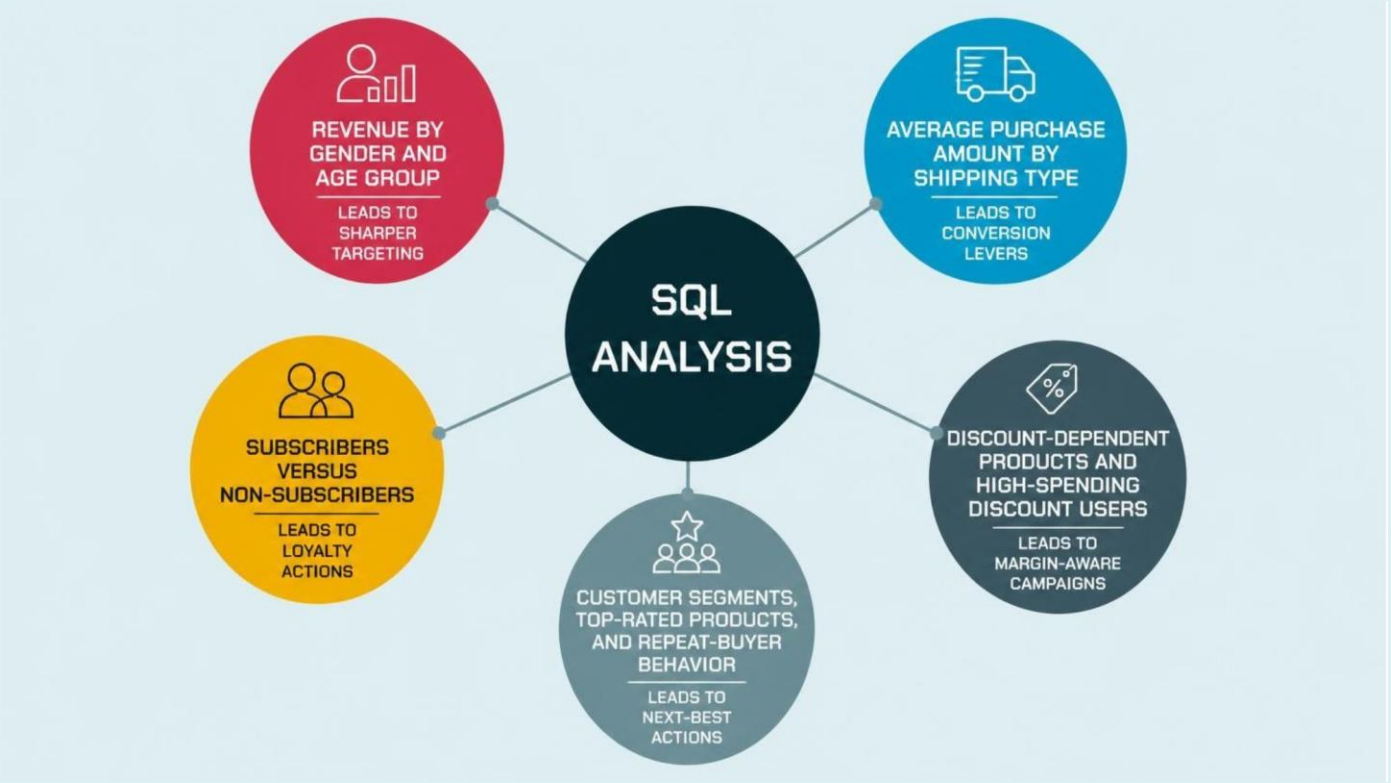
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04 · ANALYSIS

FROM QUESTION TO DECISION

SQL turned raw transactions into business questions.

SQL turned raw transactions into business questions



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Revenue analysis creates sharper targeting opportunities

COMPARISON

Gender × age group

The analysis compares revenue contribution by gender and age group.

Directional comparison; exact values remain in the source dashboard.

TARGETING SIGNAL

Male customers are the larger share of the customer base

Use the comparison to investigate differences in product preference, spending patterns, and acquisition potential.

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Products earn attention through both satisfaction and demand



SATISFACTION

Top five products by average rating

Rating leaders show where customers already express confidence.



DEMAND

Top three products within each category

Best sellers show where purchase demand is already concentrated.

Promote highly rated and best-selling products to build trust and attract attention.

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Discounts and shipping reveal conversion levers

Discount behavior

The analysis identifies products with the highest percentage of discounted purchases.

Shipping behavior

It compares average purchase amounts for Standard and Express shipping.

Campaign design

Use discount analysis to balance sales volume, average order value, and profit margin.

Audience focus

Focus campaigns on customers who prefer express shipping when the behavior supports it.



Subscriptions are a loyalty opportunity, not just a status field

01

COMPARE

Average spend and total revenue for subscribers and non-subscribers.

02

CHECK

Whether customers with more than five purchases are more likely to subscribe.

03

ACTIVATE

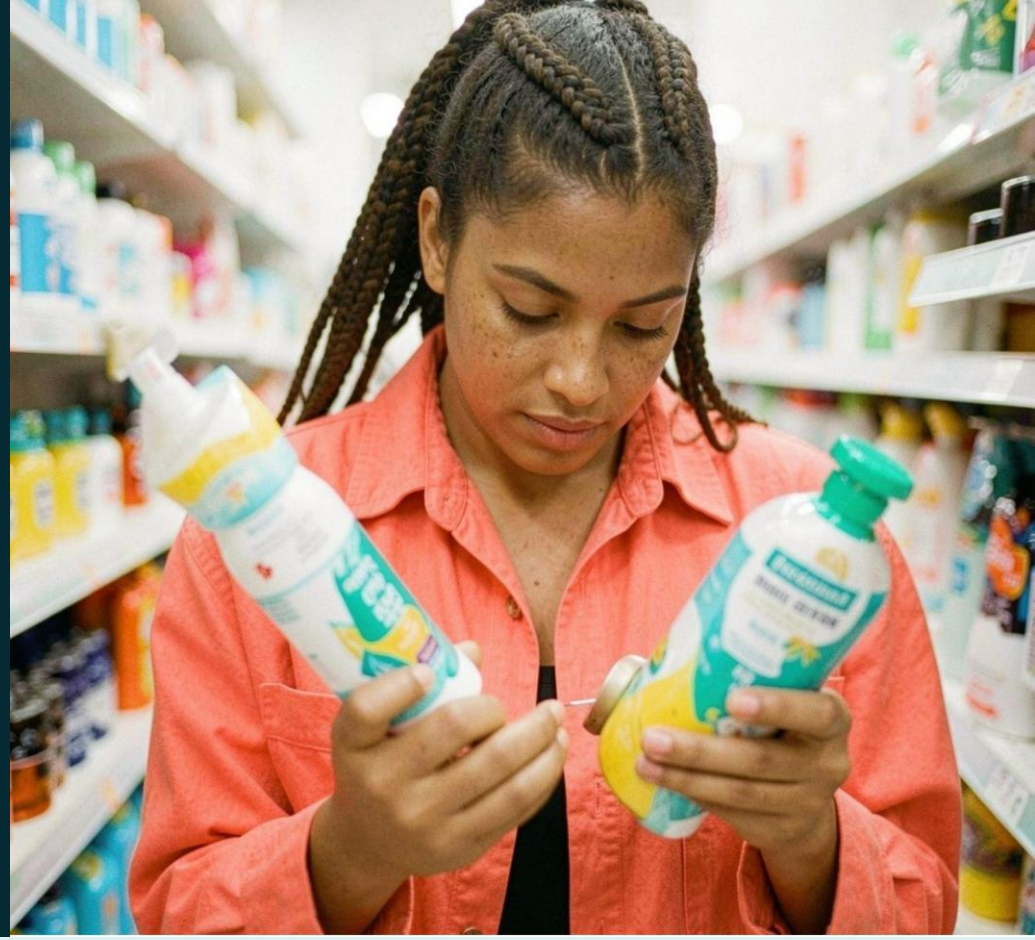
Exclusive benefits, limited-time trials, and checkout promotion.

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09 · RECOMMENDATIONS

NEXT-BEST ACTION

Segmentation turns purchase history into next-best actions.



NEW

Encourage the second purchase with a personalized offer.

RETURNING

Increase frequency through loyalty rewards.

LOYAL

Retain high-value relationships with exclusive benefits.

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